

# **Intelligenza Artificiale**

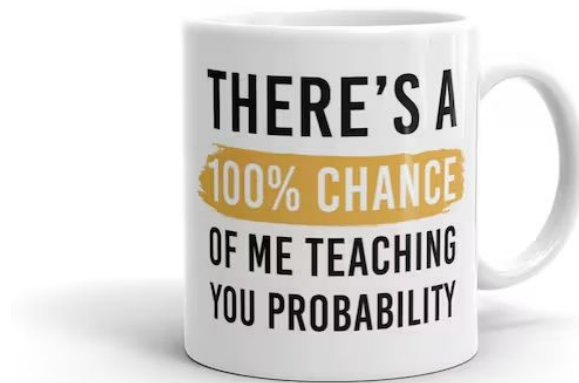
## “Quantifying uncertainty”

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# Lecture outline

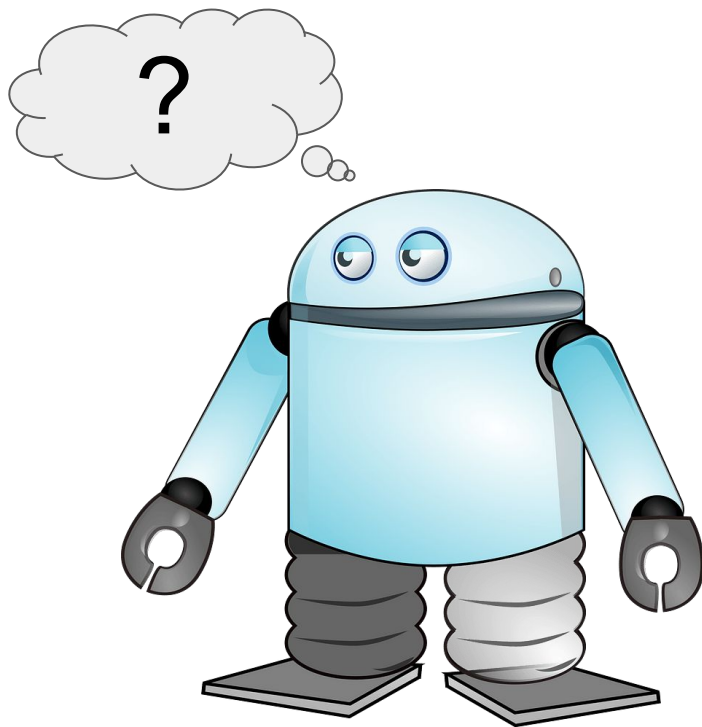
- Acting under uncertainty
- Basic probability notation
- Inference using full joint distributions
- Independence
- Bayes' rule and its use



Slides partly based on Russell & Norvig instructors material <http://aima.cs.berkeley.edu/instructors.html>

# Introduction

- We are interested in **intelligent agents** that take **rational decision** based on specific goals and the likelihood to achieve them
- **Probability theory** is a tool for representing and summarising the **uncertainty** of the agent's world
- A rational agent selects the next action based on the average of all the possible outcomes, weighted by probabilities



# Acting under uncertainty

- Real world problems contain **uncertainties due to**:
  - **partial observability**
  - **nondeterminism**
  - **adversaries**
- Example of dental diagnosis using propositional logic

*Toothache  $\Rightarrow$  Cavity*

- but not all patients with toothaches have cavities

*Toothache  $\Rightarrow$  Cavity  $\vee$  GumProblem  $\vee$  Abscess...*

- In order to make the rule true, we would have to add a long list of possible problems

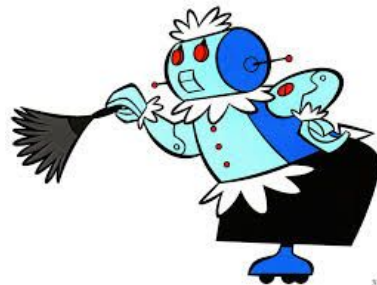


# Acting under uncertainty

- An agent's choice to do the right thing – the **rational decision** – depends on both the importance of various goals and the likelihood they will be achieved
- Large domains, such as medical diagnosis, fail for three main reasons:
  - **Laziness** – e.g. it is too much work to list the complete set of causes and symptoms needed to ensure an accurate diagnosis
  - **Theoretical ignorance** – e.g. medical science has no complete theory for the whole domain
  - **Practical ignorance** – e.g. even if we knew all the rules, we might be uncertain about a particular patient because not all the necessary tests have/can be run

# Acting under uncertainty

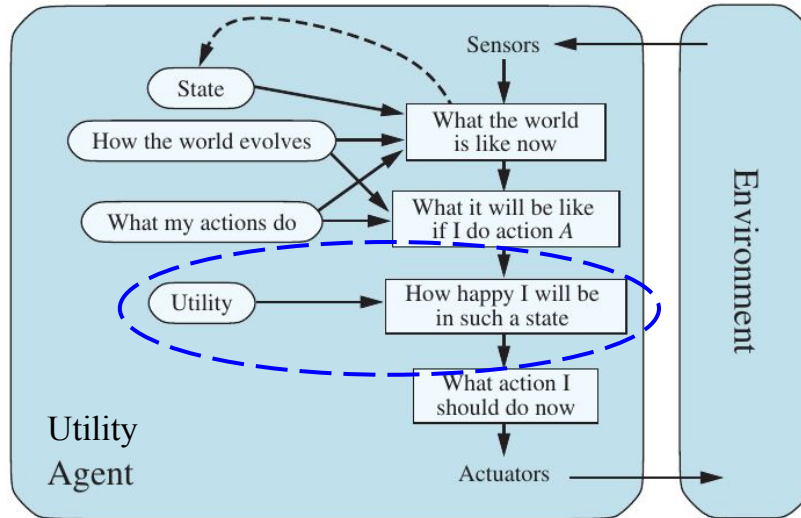
- **Probability theory**
  - tool to deal with **degrees of belief** of relevant sentences
  - summarizes the uncertainty that comes from our laziness and ignorance



# Acting under uncertainty

- How to take a rational decision under uncertainty?
  - Agent requires **preference** among **different possible outcomes** of various plans
  - **Utility Theory**: the quality of the outcome being useful
    - Every state has a degree of usefulness/utility
    - **Highest utility** is preferred
  - **Utility theory + Probability theory = Decision Theory**
    - agent is **rational** iff it chooses the action that **yields the maximum expected utility**, averaged over all the **possible outcomes** of the action

# Decision-theoretic agent



+ Probability Theory



# Decision-theoretic agent

- Function of a decision-theoretic agent that selects rational actions

**function** DT-AGENT(*percept*) **returns** an *action*

**persistent:** *belief\_state*, probabilistic beliefs about the current state of the world  
*action*, the agent's action

update *belief\_state* based on *action* and *percept*

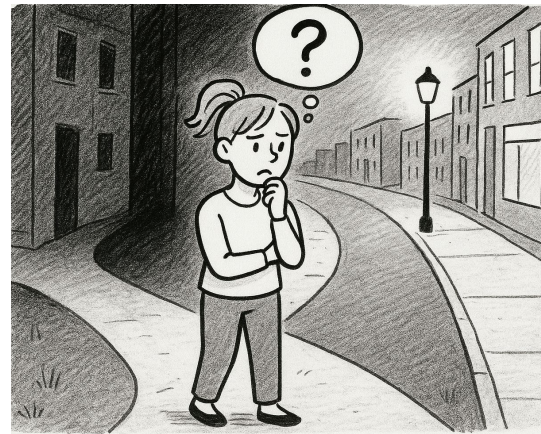
calculate outcome probabilities for actions,

given action descriptions and current *belief\_state*

select *action* with highest expected utility

given probabilities of outcomes and utility information

**return** *action*



E.g. should I take the dark shortcut on the left  
or the longer but well-lit path on the right?

# Basic probability notation

- For our agent to represent and use probabilistic information, we need a formal language
- A fully specified probability model associates a numerical **probability**  $P(\omega)$  with each possible world  $\omega \in \mathcal{W}$  (all possible worlds)
- A subset of outcomes is an **event**



# Random variables

- A **random variable** can represent any property or feature of a process with a number
- E.g. random variable that represents the possible outcomes of a die roll

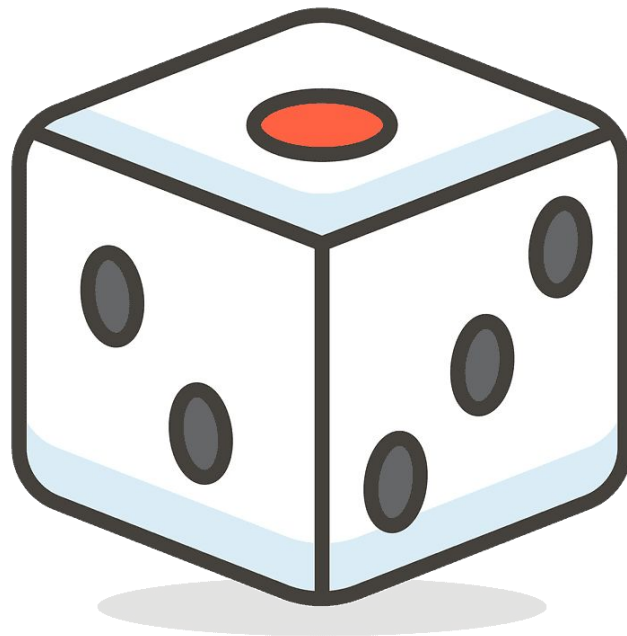
$$D \in \{1, 2, 3, 4, 5, 6\}$$

- One such outcome could be

$$D = 1$$

- In this case the **probability** of the outcome is

$$P(D = 1) = 1/6$$



# Probability axioms

- Basic axioms

$$0 \leq P(e) \leq 1$$

$$\sum_{e \in W} P(e) = 1$$

i.e. the probability is always between 0 and 1, and the sum of the probabilities of all the possible outcomes  $e$  in a given sample space  $W$  is 1

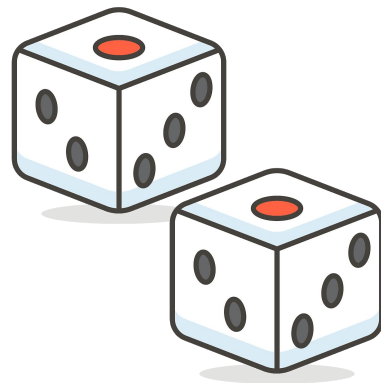
- In the die-roll case indeed

$$P(D = 1) + P(D = 2) + \dots + P(D = 6) = 1$$

# Joint probability

- We can also have multiple variables, e.g. in a two dice roll

$$P(D_1 = 1, D_2 = 1)$$



- More generally we can write the **joint probability** of multiple outcomes as

$$P(e_1, e_2, \dots, e_n) \equiv P(e_1 \wedge e_2 \wedge \dots \wedge e_n)$$

# Unconditional and Conditional probability

- The **unconditional probability** (or **prior**), for example  $P(a)$ , is the degree of belief in a proposition  $a$  in the absence of any other information
- The **conditional probability** (or **posterior**) of  $a$  “given”  $b$

$$P(a|b) = \frac{P(a, b)}{P(b)}$$

is the probability of outcome  $a$  to be true **given that** condition  $b$  is true

- E.g. probability of the sum of two dice given the outcome of the first die

$$P(D_1 + D_2 = y | D_1 = x) = \frac{P(D_1 + D_2 = y, D_1 = x)}{P(D_1 = x)}$$

# Conditional probability

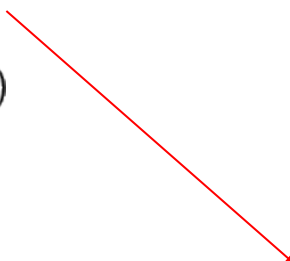
- Example

$$P(D_1 + D_2 = 7 | D_1 = 3) = ?$$

- There are  $6 \times 6 = 36$  possible outcomes, but only 1 case (i.e.  $D_2 = 4$ ) in which

$$(D_1 + D_2 = 7) \wedge (D_1 = 3)$$

- Therefore

$$P(D_1 + D_2 = 7 | D_1 = 3) = \frac{P(D_1 + D_2 = 7, D_1 = 3)}{P(D_1 = 3)} = \frac{1/36}{1/6} = \frac{1}{6}$$


1/6

0

1/3

0

1/2

0

$$P(D_1 + D_2 < 7 | D_1 = D_2) = ?$$

None of the above

0





# Conditional probability

- Example

$$P(D_1 + D_2 < 7 | D_1 = D_2) = ?$$

- Still 36 possible outcomes, but now 3 cases (1-1, 2-2 and 3-3) satisfying

$$(D_1 + D_2 < 7) \wedge (D_1 = D_2)$$

- Therefore

$$P(D_1 + D_2 < 7 | D_1 = D_2) = \frac{P(D_1 + D_2 < 7, D_1 = D_2)}{P(D_1 = D_2)} = \frac{3/36}{6/36} = \frac{1}{2}$$

# Independence

- Two events  $a$  and  $b$  are **independent** if

$$P(a|b) = P(a)$$

- or equivalently

$$P(b|a) = P(b) \quad \text{or} \quad P(a, b) = P(a)P(b)$$

- For example

- $P(D_2 = 2|D_1 = 1) = P(D_2 = 2) = \frac{1}{6}$

- $P(D_1 = 1, D_2 = 2) = P(D_1 = 1)P(D_2 = 2) = \frac{1}{36}$

# Conditional independence

- Two events  $a$  and  $b$  are **conditionally independent** given a third event  $c$  if

$$P(a|b, c) = P(a|c)$$

- or equivalently

$$P(b|a, c) = P(b|c) \quad \text{or} \quad P(a, b|c) = P(a|c)P(b|c)$$

- E.g. for a smoke alarm device



$$P(\text{alarm}|\text{fire}, \text{smoke}) = P(\text{alarm}|\text{smoke})$$

# Law of total probability

- If  $\{b_1, b_2, \dots, b_N\}$  is a set of mutually exclusive events (i.e. only one event at a time can be true), the **law of total probability** states

$$P(a) = P(a, b_1) + P(a, b_2) + \dots + P(a, b_N) = \sum_{k=1}^N P(a, b_k)$$

- E.g. in a two-dice roll

$$P(D_1 = 1) = P(D_1 = 1, D_2 = 1) + \dots + P(D_1 = 1, D_2 = 6)$$

- We can also write

$$P(a) = \sum_{k=1}^N P(a, b_k) = \sum_{k=1}^N P(a|b_k)P(b_k)$$

# Probability distribution

- The **probability distribution**  $\mathbf{P}(X)$  is the **set of probabilities** assigned to all the possible values of the random variable  $X$

$$\mathbf{P}(X) = \{P(x_1), P(x_2), P(x_3), \dots\}$$

$$X \in \{x_1, x_2, x_3, \dots\} \qquad \sum_{x_i} P(x_i) = 1$$

- For example the die-roll has a (**uniform**) probability distribution

$$\mathbf{P}(D) = \left\{ \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6} \right\}$$

# Probability distribution

- Example: the statement **P** (different from “single” probability *P*) defines a probability distribution for the random variable *Weather*

$$P(\text{Weather} = \text{sun}) = 0.6$$

$$P(\text{Weather} = \text{rain}) = 0.1$$

$$P(\text{Weather} = \text{cloud}) = 0.29$$

$$P(\text{Weather} = \text{snow}) = 0.01$$

- Possible representations:

$$\mathbf{P}(\text{Weather}) = \{0.6, 0.1, 0.29, 0.01\} \text{ or } (0.6, 0.1, 0.29, 0.01) \text{ or } \langle 0.6, 0.1, 0.29, 0.01 \rangle$$

# Probability density function

- For a continuous random variable, the distribution is substituted by a **probability density function (pdf)**

$$p(X = x) \equiv p(x)$$

$$\int p(x) dx = 1$$

- This distinction is due to the infinite number of possible values of a continuous variable – each one with infinitesimally small probability
- In this course we will mostly deal with discrete random variables

# Inference using full joint distributions

- **Example:** the dentist touches one of my teeth, and it hurts! :-(
  - I want to understand whether the pain is due to
    - 1) a toothache                      2) a cavity                      3) the dentist's tool catching my tooth
  - The **full joint distribution** of the 3 boolean variables,  $\mathbf{P(Toothache, Cavity, Catch)}$ , can be represented by a table with  $2^3$  probabilities
    - Being a distribution, the sum of these probabilities must be 1

|                      | <i>toothache</i> |                     | $\neg$ <i>toothache</i> |                     |
|----------------------|------------------|---------------------|-------------------------|---------------------|
|                      | <i>catch</i>     | $\neg$ <i>catch</i> | <i>catch</i>            | $\neg$ <i>catch</i> |
| <i>cavity</i>        | .108             | .012                | .072                    | .008                |
| $\neg$ <i>cavity</i> | .016             | .064                | .144                    | .576                |





# Inference using full joint distributions

- Start with the joint distribution:

|                      | <i>toothache</i> |                     | $\neg$ <i>toothache</i> |                     |
|----------------------|------------------|---------------------|-------------------------|---------------------|
|                      | <i>catch</i>     | $\neg$ <i>catch</i> | <i>catch</i>            | $\neg$ <i>catch</i> |
| <i>cavity</i>        | .108             | .012                | .072                    | .008                |
| $\neg$ <i>cavity</i> | .016             | .064                | .144                    | .576                |

- For any proposition  $\phi$ , sum the probabilities of atomic events  $\omega$  for which  $\phi$  is true:

$$P(\phi) = \sum_{\omega: \omega \models \phi} P(\omega)$$

$$P(\text{toothache}) = 0.108 + 0.012 + 0.016 + 0.064 = \mathbf{0.2}$$

# Inference using full joint distributions

|                      | <i>toothache</i> |                     | $\neg$ <i>toothache</i> |                     |
|----------------------|------------------|---------------------|-------------------------|---------------------|
|                      | <i>catch</i>     | $\neg$ <i>catch</i> | <i>catch</i>            | $\neg$ <i>catch</i> |
| <i>cavity</i>        | .108             | .012                | .072                    | .008                |
| $\neg$ <i>cavity</i> | .016             | .064                | .144                    | .576                |

$$P(\text{cavity} \vee \text{toothache}) = 0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064 = 0.28$$

# Inference using full joint distributions

|                      | <i>toothache</i> |                     | $\neg$ <i>toothache</i> |                     |
|----------------------|------------------|---------------------|-------------------------|---------------------|
|                      | <i>catch</i>     | $\neg$ <i>catch</i> | <i>catch</i>            | $\neg$ <i>catch</i> |
| <i>cavity</i>        | .108             | .012                | .072                    | .008                |
| $\neg$ <i>cavity</i> | .016             | .064                | .144                    | .576                |

$$\begin{aligned}P(\neg \text{cavity} \mid \text{toothache}) &= \frac{P(\neg \text{cavity} \wedge \text{toothache})}{P(\text{toothache})} \\&= (0.016 + 0.064) / (0.108 + 0.012 + 0.016 + 0.064) \\&= \mathbf{0.4}\end{aligned}$$

# Normalization

- In the last example, we can consider both cases of *cavity* and  $\neg$ *cavity* in case of *toothache* writing the probability distribution  $\mathbf{P}(\text{Cavity} \mid \text{toothache})$

$$\mathbf{P}(\text{Cavity} \mid \text{toothache}) = \mathbf{P}(\text{Cavity}, \text{toothache}) / P(\text{toothache})$$

$$\begin{aligned}\Rightarrow \sum_{\text{Cavity}} \mathbf{P}(\text{Cavity} \mid \text{toothache}) &= \sum_{\text{Cavity}} \mathbf{P}(\text{Cavity}, \text{toothache}) / P(\text{toothache}) \\ &= [1 / P(\text{toothache})] \sum_{\text{Cavity}} \mathbf{P}(\text{Cavity}, \text{toothache}) \\ &= \alpha \sum_{\text{Cavity}} \mathbf{P}(\text{Cavity}, \text{toothache}) = 1\end{aligned}$$

- The term  $\alpha = 1 / P(\text{toothache})$  can be viewed as a **normalization constant** so that the total probability of the distribution  $\mathbf{P}(\text{Cavity} \mid \text{toothache})$  is 1

# Normalization – numerical example

|                      | <i>toothache</i> |                     | $\neg$ <i>toothache</i> |                     |
|----------------------|------------------|---------------------|-------------------------|---------------------|
|                      | <i>catch</i>     | $\neg$ <i>catch</i> | <i>catch</i>            | $\neg$ <i>catch</i> |
| <i>cavity</i>        | <b>.108</b>      | <b>.012</b>         | .072                    | .008                |
| $\neg$ <i>cavity</i> | <b>.016</b>      | <b>.064</b>         | .144                    | .576                |

- We could apply the law of total probability considering the values of the *Catch* variable:

$$\begin{aligned}
 \mathbf{P(Cavity \mid toothache)} &= \alpha \mathbf{P(Cavity, toothache)} \\
 &= \alpha [\mathbf{P(Cavity, toothache, catch)} + \mathbf{P(Cavity, toothache, \neg catch)}] \\
 &= \alpha [<\mathbf{0.108, 0.016}> + <\mathbf{0.012, 0.064}>] \\
 &= \alpha <0.12, 0.08> \\
 &= <\mathbf{0.6, 0.4}>
 \end{aligned}$$

with  $\alpha = 1 / (0.12 + 0.08) = 5$



# Inference with marginalisation

- Let  $\mathbf{X}$  be **all** the variables. Typically, we want
  - the **posterior** joint distribution of the **query** variables  $\mathbf{Y}$ ...
  - ... given specific values  $\mathbf{e}$  for the **evidence** variables  $\mathbf{E}$
- Let the **hidden** variables be  $\mathbf{H} = \mathbf{X} - \mathbf{Y} - \mathbf{E}$
- To answer query  $\mathbf{Y}$  given evidence  $\mathbf{e}$  we can sum out (**marginalise**) the hidden variables as follows

$$\begin{aligned}\mathbf{P}(\mathbf{Y} \mid \mathbf{E} = \mathbf{e}) &= \alpha \mathbf{P}(\mathbf{Y}, \mathbf{E} = \mathbf{e}) && \text{apply law of total probability} \\ &= \alpha \sum_{\mathbf{h}} \mathbf{P}(\mathbf{Y}, \mathbf{E} = \mathbf{e}, \mathbf{H} = \mathbf{h}) && \text{then normalize}\end{aligned}$$

- E.g. in the previous slide  $\mathbf{Y} = \text{Cavity}$ ,  $\mathbf{e} = \text{toothache}$ , and  $\mathbf{H} = \text{Catch}$

# Product and Bayes Rules

- From the definition of conditional probability, we can write the **product rule**

$$P(a, b) = P(a|b)P(b)$$

- But because  $P(a, b) = P(b, a)$ , we can also write the **Bayes rule**

$$P(a|b)P(b) = P(b|a)P(a)$$

- Another way of writing the same

$$P(a|b) = \frac{P(b|a)}{P(b)} P(a)$$

# Use of Bayes rule

- Often, we perceive as evidence the **effect** of some unknown **cause**, and we would like to determine the latter – in that case, Bayes' rule becomes

$$P(\text{cause} \mid \text{effect}) = \frac{P(\text{effect} \mid \text{cause})}{P(\text{effect})} P(\text{cause})$$

- The conditional probability  $P(\text{effect} \mid \text{cause})$  quantifies the relationship in the **causal** direction, whereas  $P(\text{cause} \mid \text{effect})$  describes the **diagnostic** direction



# Example of Bayes rule's application

- A doctor knows that meningitis causes a patient to have a stiff neck 70% of the time. The doctor also knows the prior (unconditional) probability that any patient has meningitis is 1/50000, and the prior that any patient has a stiff neck is 1%. If  $c$  = “meningitis” (**cause**) and  $e$  = “stiff neck” (**effect**)

$$P(e|c) = 0.7$$

$$P(c) = 1/50000$$

$$P(e) = 0.01 \gg P(c)$$

$$P(c|e) = P(e|c) P(c) / P(e) = (0.7 \times 1/50000) / 0.01 = \mathbf{0.0014}$$



- Only 0.14% of patients with a stiff neck have meningitis. Even though a stiff neck is very often a symptom of meningitis (with probability 0.7), the prior of stiff necks (from any cause) is much higher than the prior for meningitis  $\Rightarrow$  the probability of meningitis in patients with stiff necks remains small!

## Example (from Pearl & Mackenzie's "The Book of Why")

- Probability of 40 years old woman of getting breast cancer within a year

$$P(c) = 1/700 \quad (\sim 0.14\%)$$

- Probability of positive mammogram test given that one has cancer (sensitivity)

$$P(m|c) = 73\%$$

- Probability of positive mammogram test given that one doesn't have cancer (false positive rate)

$$P(m|\neg c) = 12\%$$

## Example (from Pearl & Mackenzie's "The Book of Why")

- **QUERY:** What is the probability of breast cancer given a positive test?

$$P(c|m) = ?$$

- (marginal.) 
$$P(m) = P(m|c)P(c) + P(m|\neg c)P(\neg c)$$
$$= 73\% \times (1/700) + 12\% \times (699/700) \approx 12.1\%$$

- (Bayes) 
$$P(c|m) = \frac{P(m|c)}{P(m)} P(c)$$
$$= (73\%/12.1\%) \times (1/700) \approx 0.9\%$$

# Summary



- **Probabilities** express the agent's inability to reach a definite decision regarding the truth of a sentence
- **Decision theory** combines the agent's beliefs and desires, defining the best action as the one that maximizes expected utility
- Basic probability statements include **prior** or **unconditional probabilities** and **posterior** or **conditional probabilities** over simple and complex propositions
- The **full joint probability distribution** specifies the probability of each complete assignment of values to random variables
- **Bayes' rule** allows unknown probabilities to be computed from known conditional probabilities, usually in the causal direction

# Conclusions

- Suggested reading
  - Chapter 12, sec. 12.1-12.5 of Russell & Norvig
  - Appendix A.3 of Russell & Norvig
  - Chapter 1, sec 1.3.1-1.3.9 of Pearl et al.
- Next lecture  
Probabilistic reasoning
- Questions?

